Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION – 2025

Name of the Course: B.Tech.-Aerospace Engineering

Semester: 4th

Paper Code: TAS-401

Name of the Paper: Academic Report writing and IPR

Time: 3 Hours

Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1		(20marks)
(a)	i) List and briefly describe three common types of academic reports used in engineering. ii) Discuss three key characteristics of academic writing that make it suitable for engineering communication.	CO1
(b)	How does the audience influence the tone, language, and structure of an engineering report? Provide examples.	CO2
(c)	With the help of a suitable figure outline the standard structure of an academic report, emphasizing its essential components.	CO2
Q2		(20 marks)
(a)	What is a citation, and why is a person's citation score significant? What do you understand by self-citation?	CO2
(b)	What do you mean by research ethics, and why are they essential in academic and scientific research?	CO2
(c)	Briefly explain the following: i) Scopus Indexed Journals ii) Journal Impact Factor	CO3
Q3		(20 marks)
(a)	What is a patent, and what are the key criteria for an invention to be patentable? Discuss the rights granted to a patent holder and the duration of patent protection.	CO3
(b)	Explain the concept of copyright and its significance in creative and academic works. How does copyright protection differ from patent protection?	CO3
(c)	What are the common legal and ethical challenges related to Intellectual Property Rights (IPR)? Provide examples of IPR infringement and their consequences.	CO3
Q4		(20 marks)
(a)	Explain the step-by-step procedure for patent filing. Discuss the essential documents needed and the significance of the patent examination process in	CO4
(b)	Discuss a famous case of Intellectual Property Rights (IPR) infringement in the aerospace sector. What were the key legal and ethical issues involved, and how was the dispute resolved?	CO4
(c)	Differentiate between patents and copyrights. Discuss the procedure for obtaining a copyright and its significance in protecting engineering-related software and technical documentation.	CO4

Q5		(20 marks)
(a)	Describe the fundamental principles of effective communication in a professional environment. What strategies can engineers and technical professionals use to	CO5, CO6
(b)	Outline the key steps in preparing a technical presentation. How can presenters structure their content to ensure clarity, and audience engagement?	CO 5, CO6
(c)	What are the best techniques for delivering a technical presentation successfully? Explain the significance of body language, voice modulation, and audience interaction in effective communication.	CO6